LEMON Open lab

2026.11.06

Contents

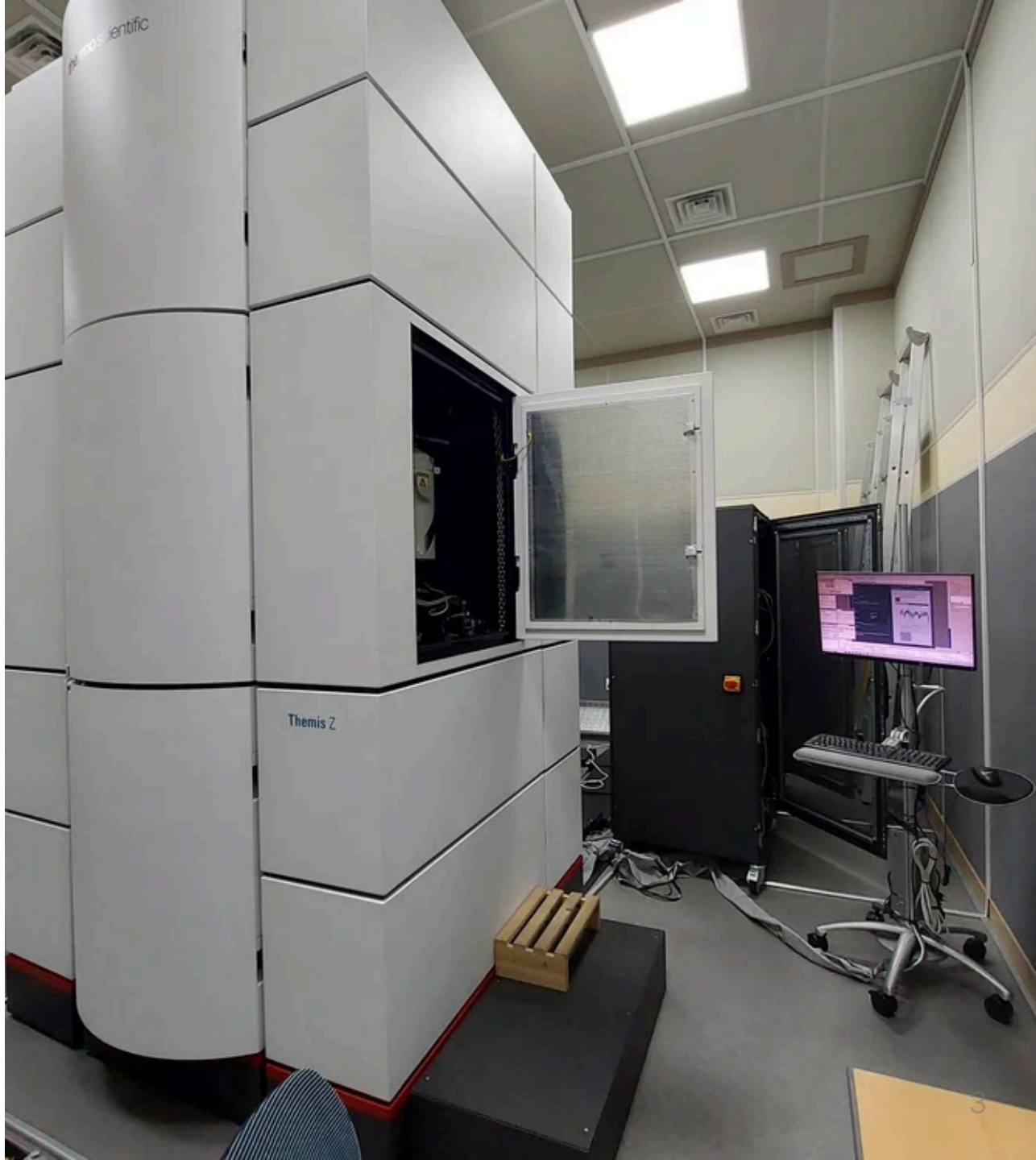
0. Lab + PLD 투어

1. TEM 소개

2. 현재 연구 소개

(Scanning) Transmission Electron Microscopes

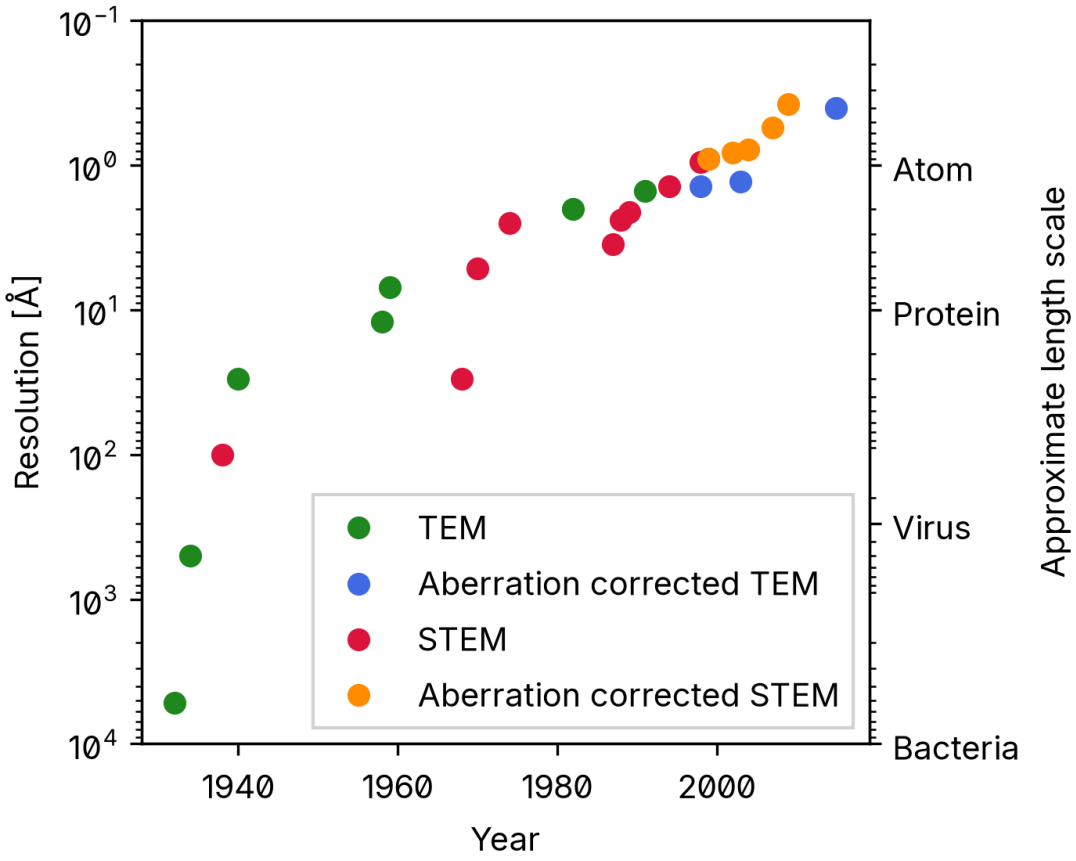
서울대학교 신소재공동연구소 Themis-Z, ...



(Scanning) Transmission Electron Microscopes

Cornell Spectra 300, ...

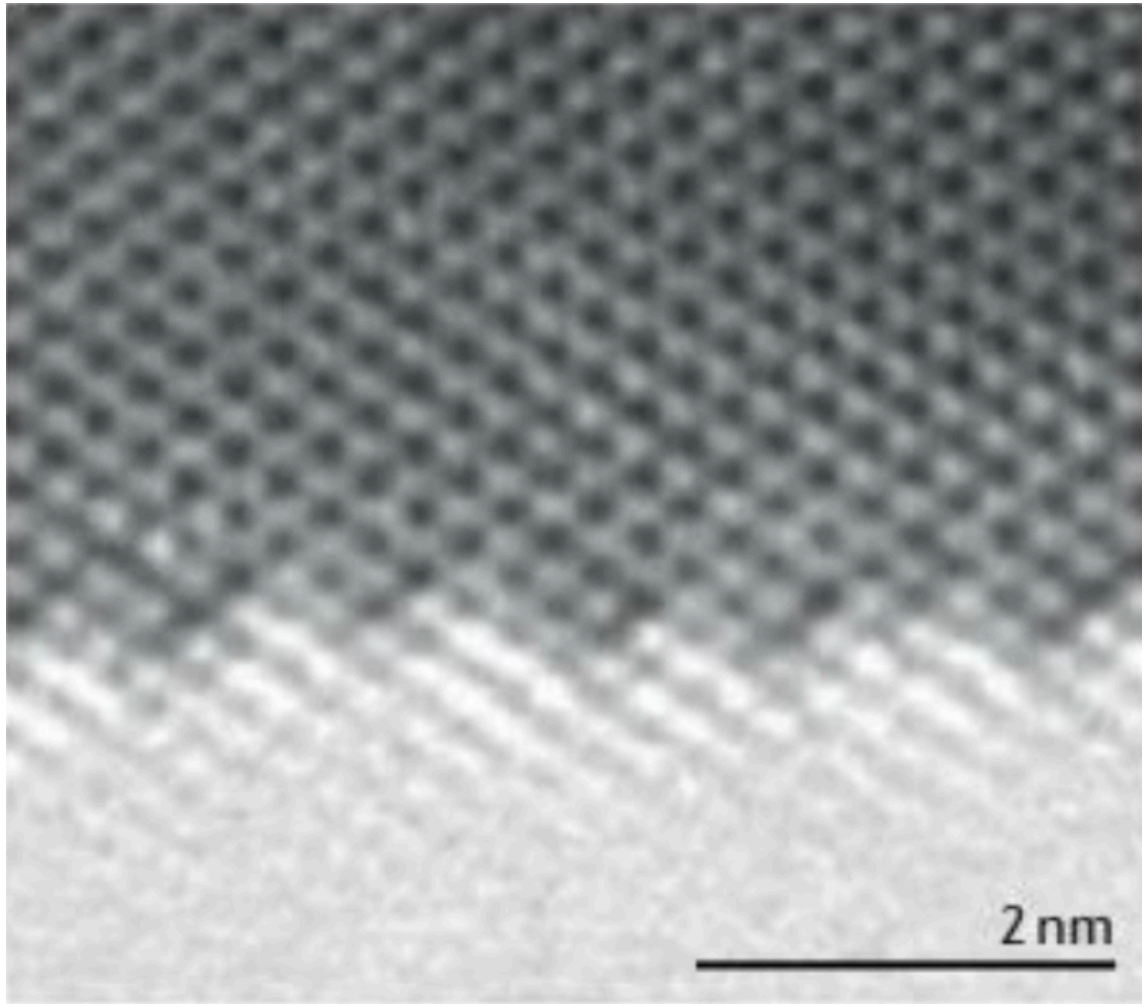
Transmission Electron Microscopy



Atomic resolution microscopy: 1~0.1 Å

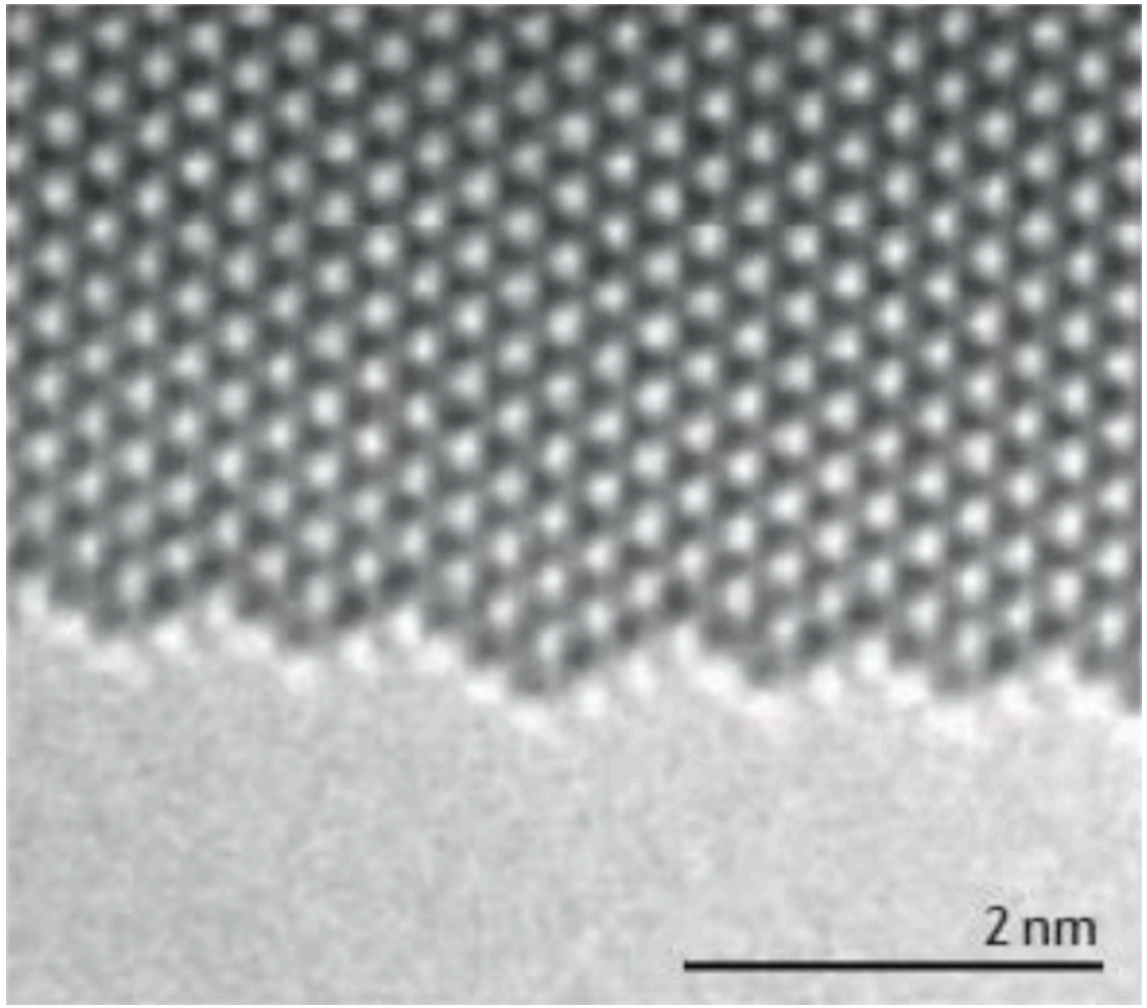
Adapted from Oxley, M.P., Lupini, A.R., Pennycook, S.J., 2016. Ultra-high resolution electron microscopy. Rep. Prog. Phys. 80, 026101. https://doi.org/10.1088/1361-6633/80/2/026101

Conventional TEM (CTEM)



de Jonge, N., Houben, L., Dunin-Borkowski, R.E., Ross, F.M., 2019. Resolution and aberration correction in liquid cell transmission electron microscopy. Nat Rev Mater 4, 61–78. https://doi.org/10.1038/s41578-018-0071-2

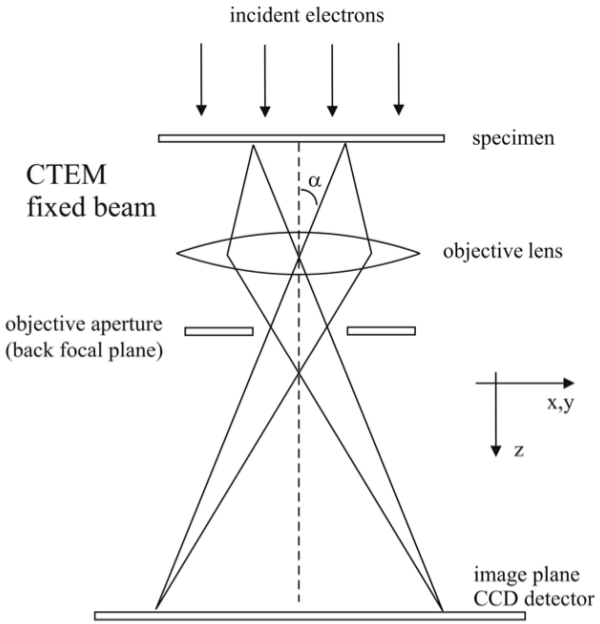
Cs-Corrected CTEM



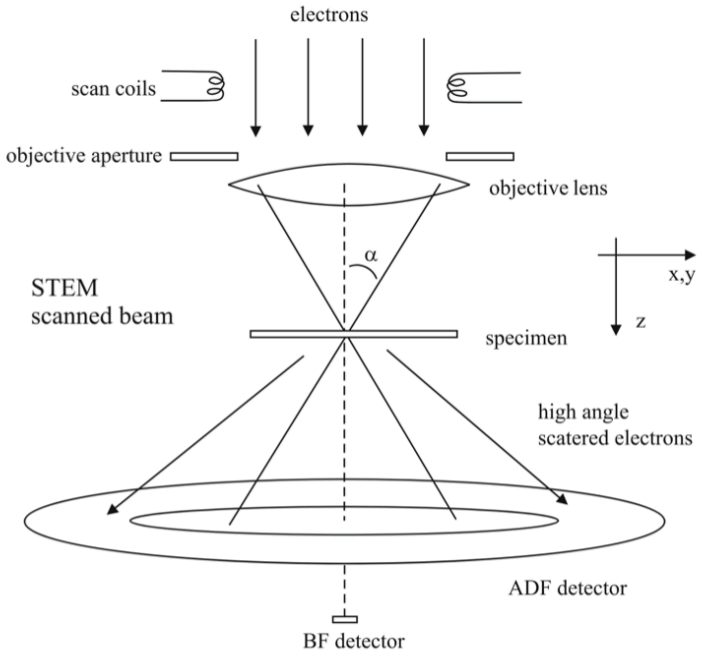
de Jonge, N., Houben, L., Dunin-Borkowski, R.E., Ross, F.M., 2019. Resolution and aberration correction in liquid cell transmission electron microscopy. Nat Rev Mater 4, 61–78. https://doi.org/10.1038/s41578-018-0071-2

Scanning TEM (STEM)

- 수차, 시편 두께, 초점 거리 등에 덜 민감 → 고해상도 이미지
- Z-contrast, strain, chemical analysis 등 새로운 분석

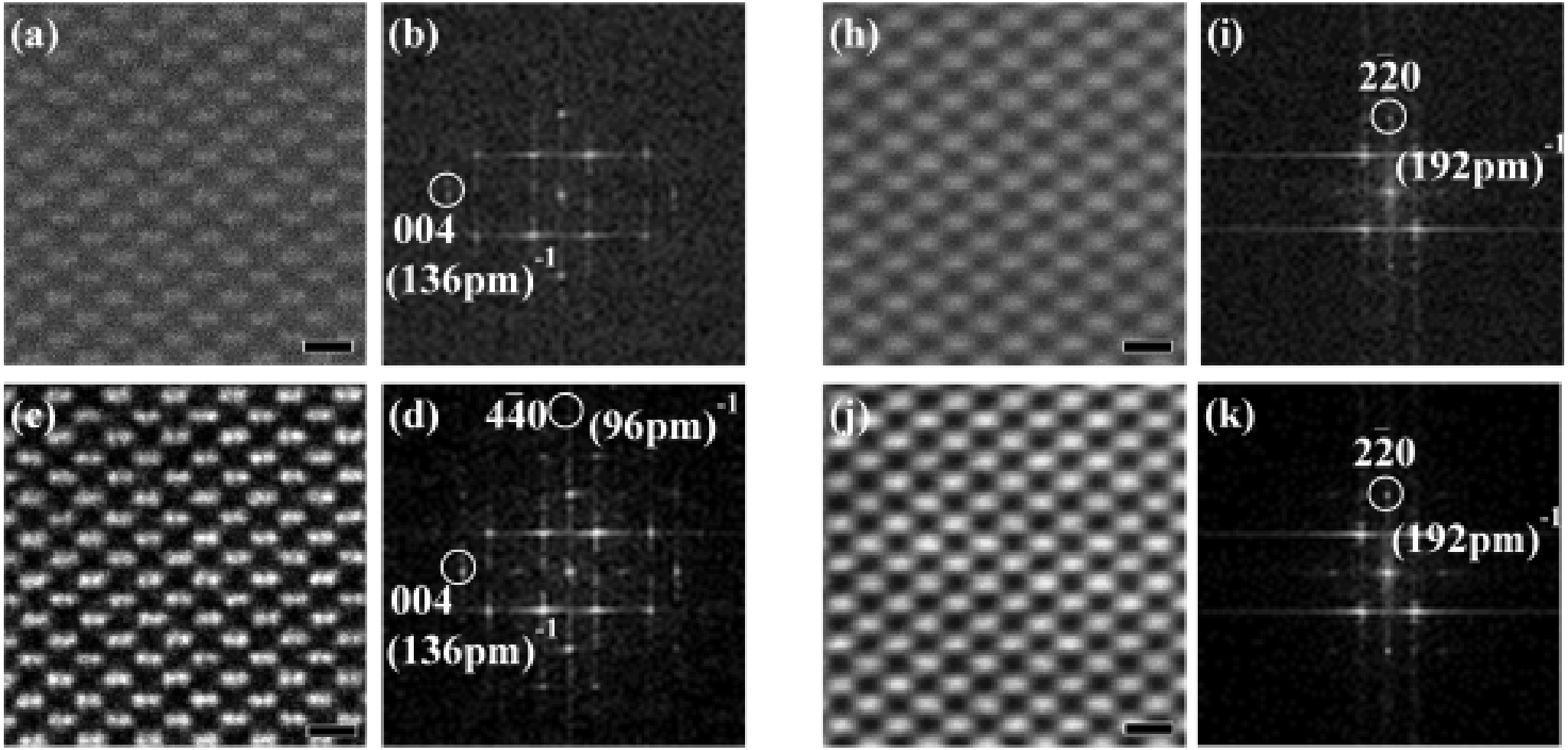


← CTEM



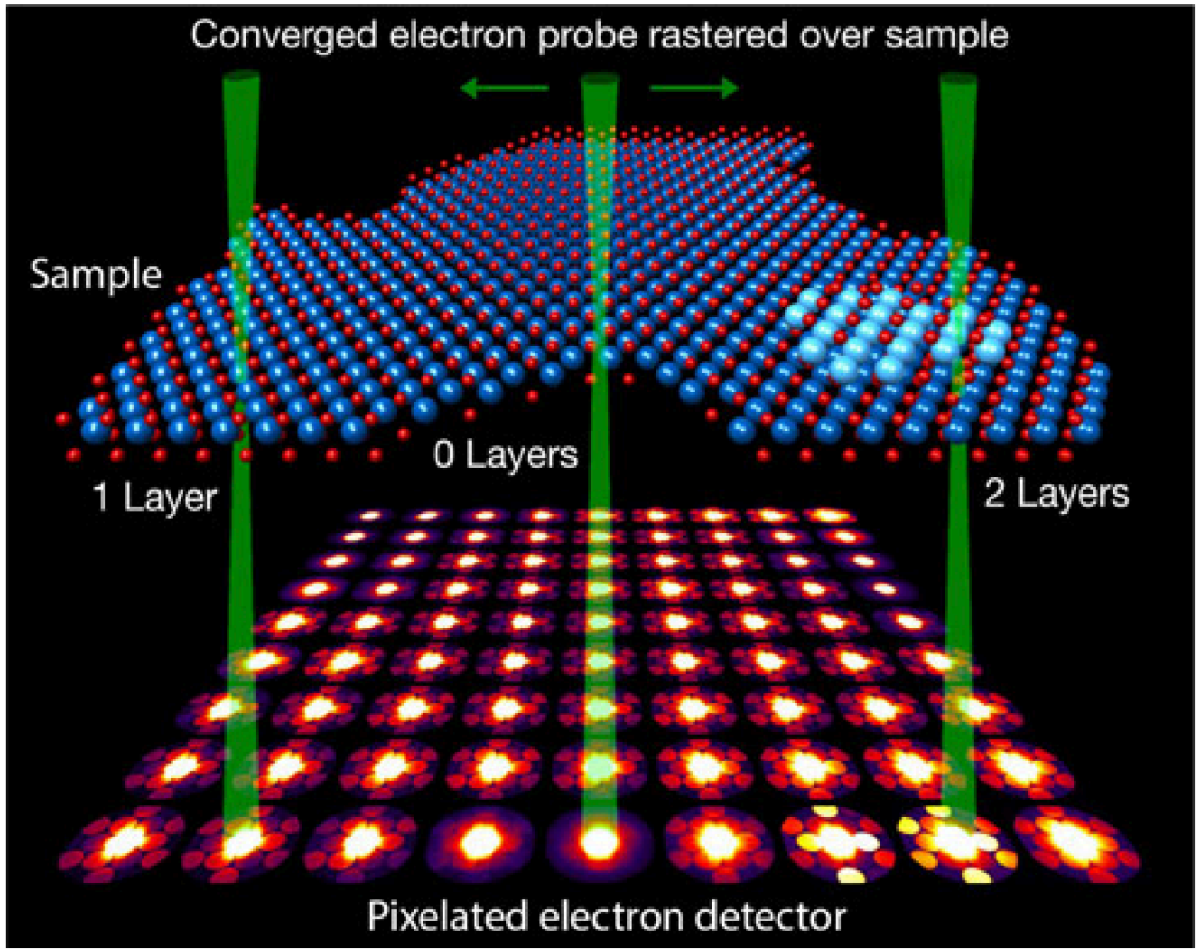
← STEM

Cs-Corrected STEM



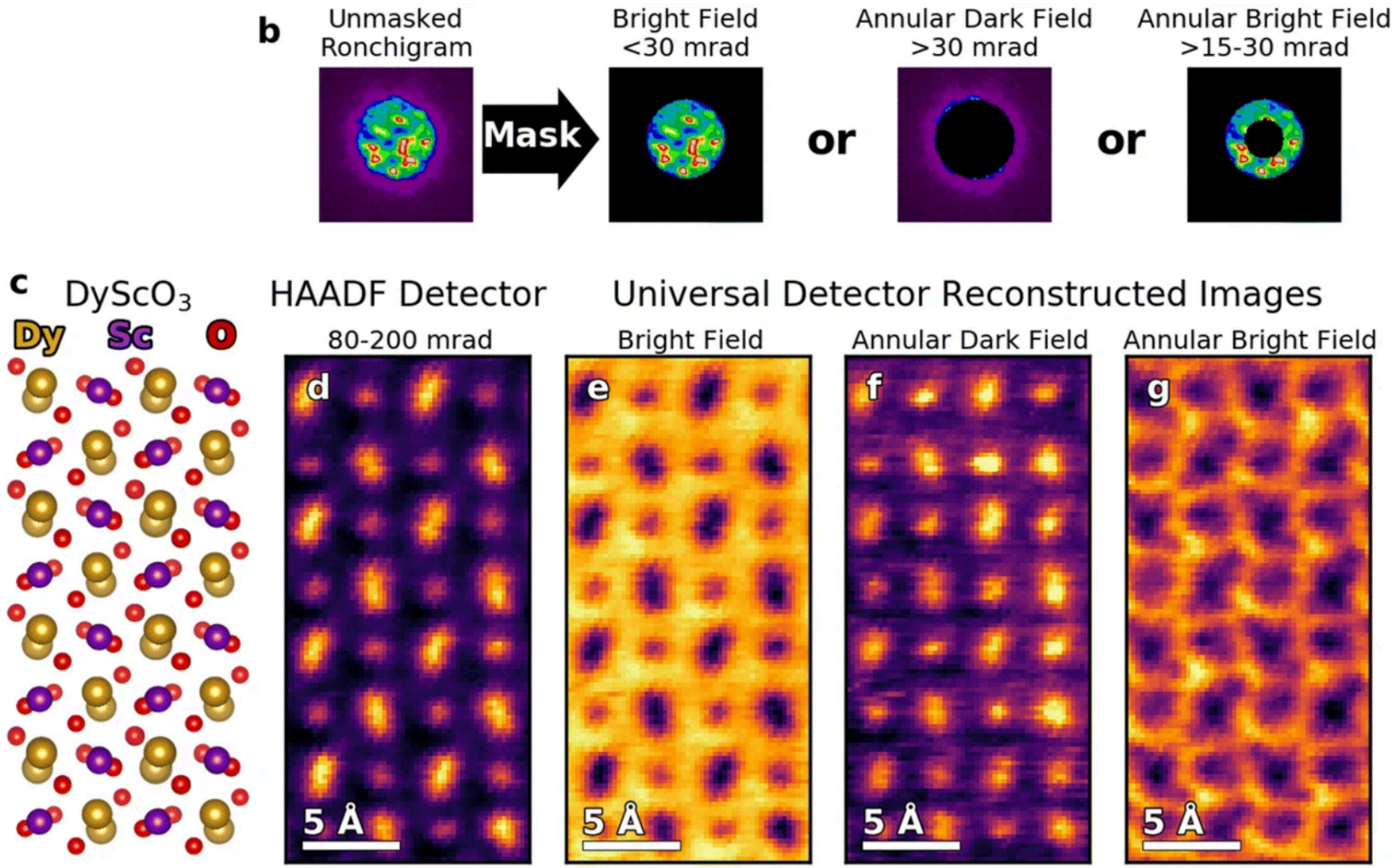
Shiloh, R. et al., 2018. Spherical aberration correction in a scanning transmission electron microscope using a sculpted thin film. Ultramicroscopy 189, 46–53. https://doi.org/10.1016/j.ultramic.2018.03.016

4D-STEM



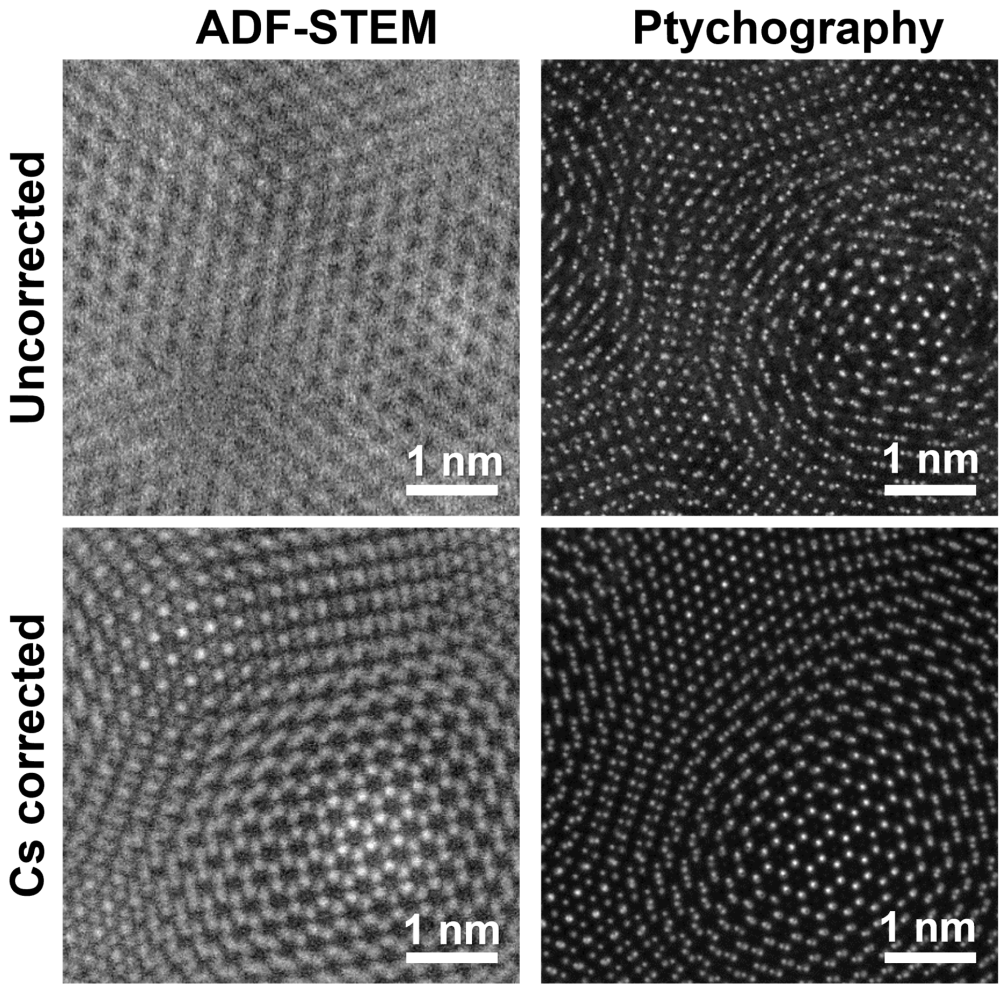
Ophus, C., 2019. Four-Dimensional Scanning Transmission Electron Microscopy (4D-STEM): From Scanning Nanodiffraction to Ptychography and Beyond. Microscopy and Microanalysis 25, 563–582. https://doi.org/10.1017/S1431927619000497

4D-STEM



Hachtel, J.A., Idrobo, J.C., Chi, M., 2018. Sub-Ångstrom electric field measurements on a universal detector in a scanning transmission electron microscope. Adv Struct Chem Imag 4, 10. https://doi.org/10.1186/s40679-018-0059-4

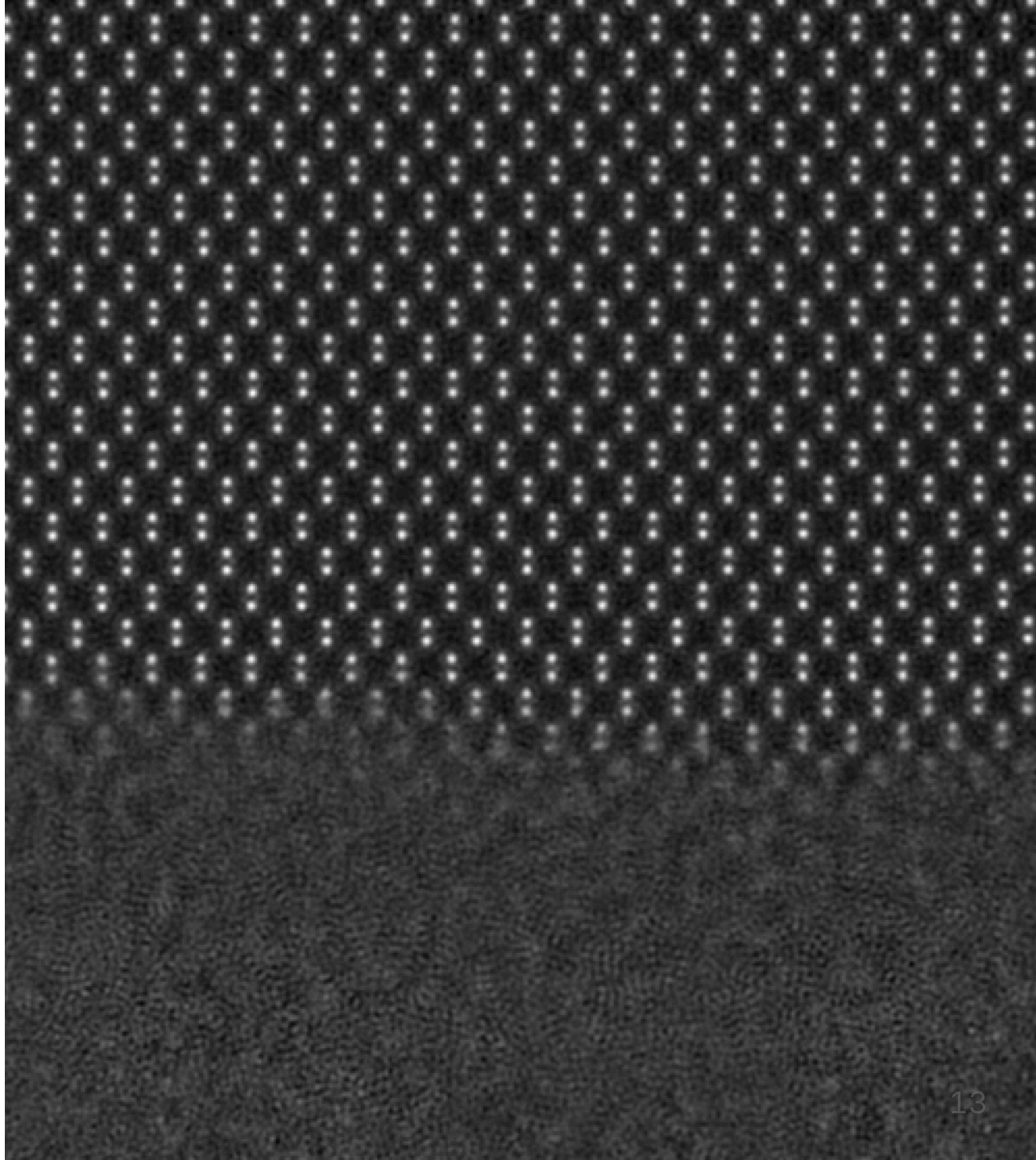
Ptychography



Nguyen, K.X., Jiang, Y., Lee, C.-H., Kharel, P., Zhang, Y., van der Zande, A.M., Huang, P.Y., 2024. Achieving sub-0.5-angstrom–resolution ptychography in an uncorrected electron microscope. Science 383, 865–870. https://doi.org/10.1126/science.adl2029

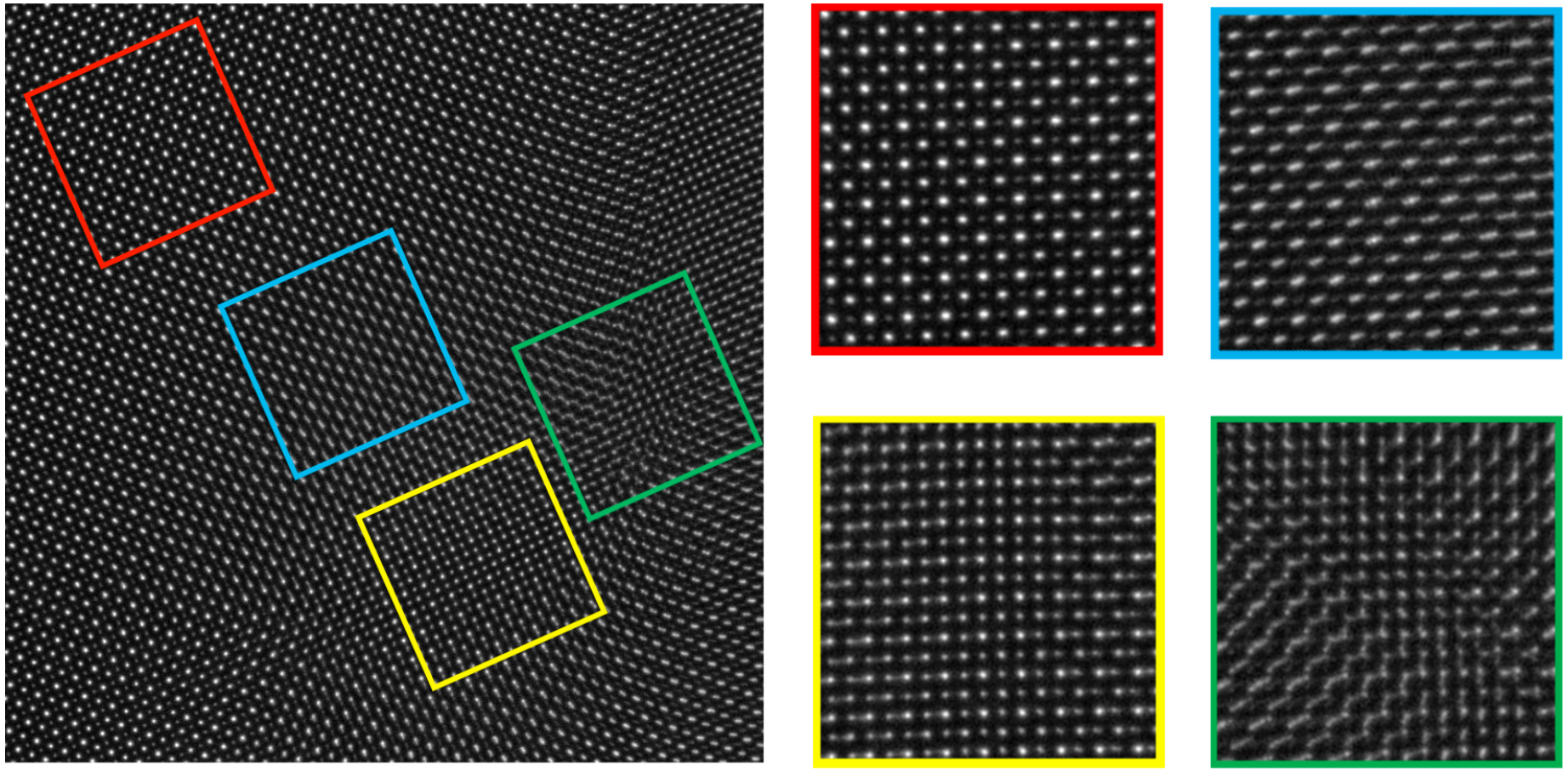
Examples

Depth-resolved 3D Ptychographic imaging of FIB-damaged Silicon



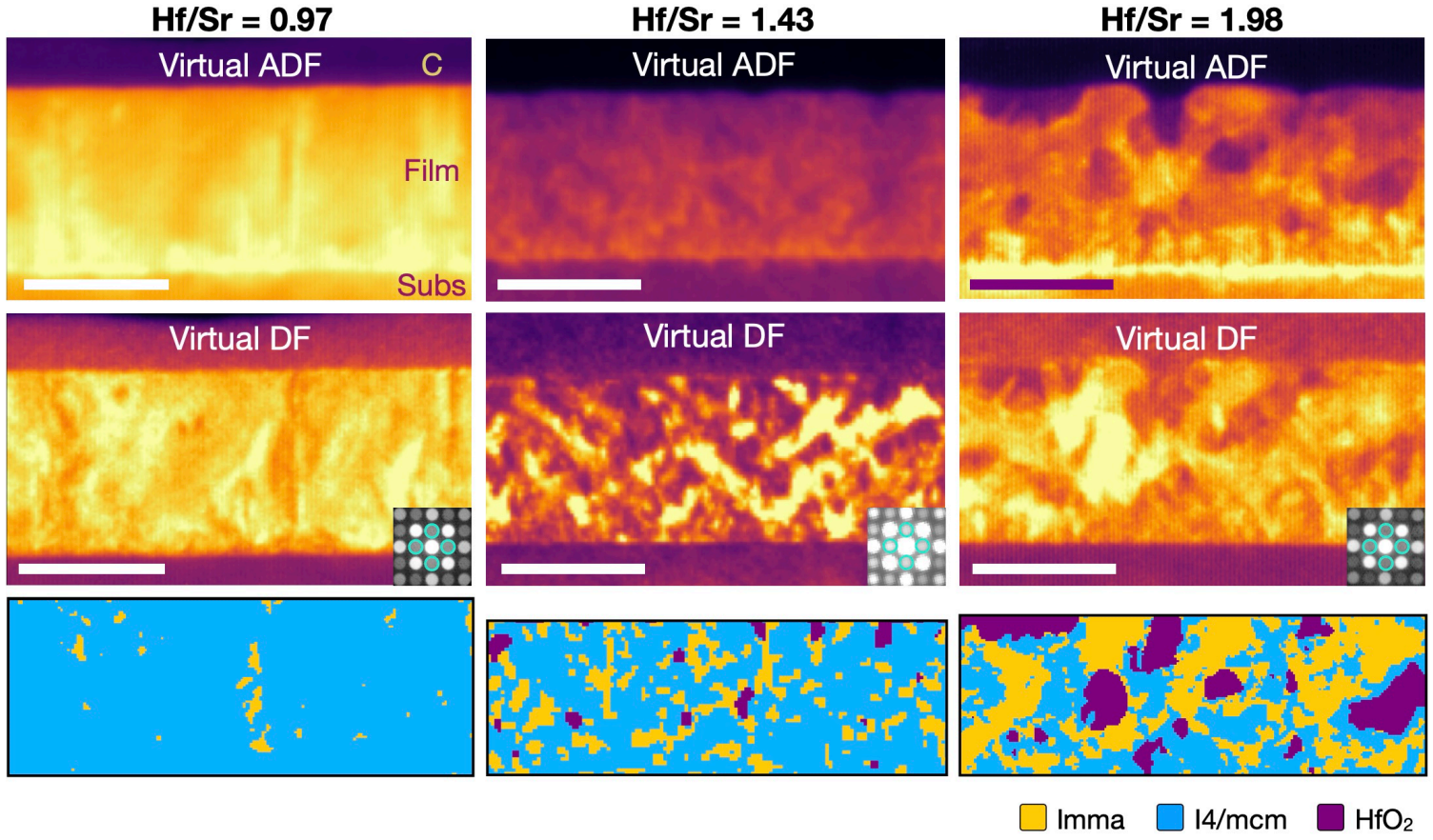
Examples

BaTiO3 dislocation analysis using 3D ptychography



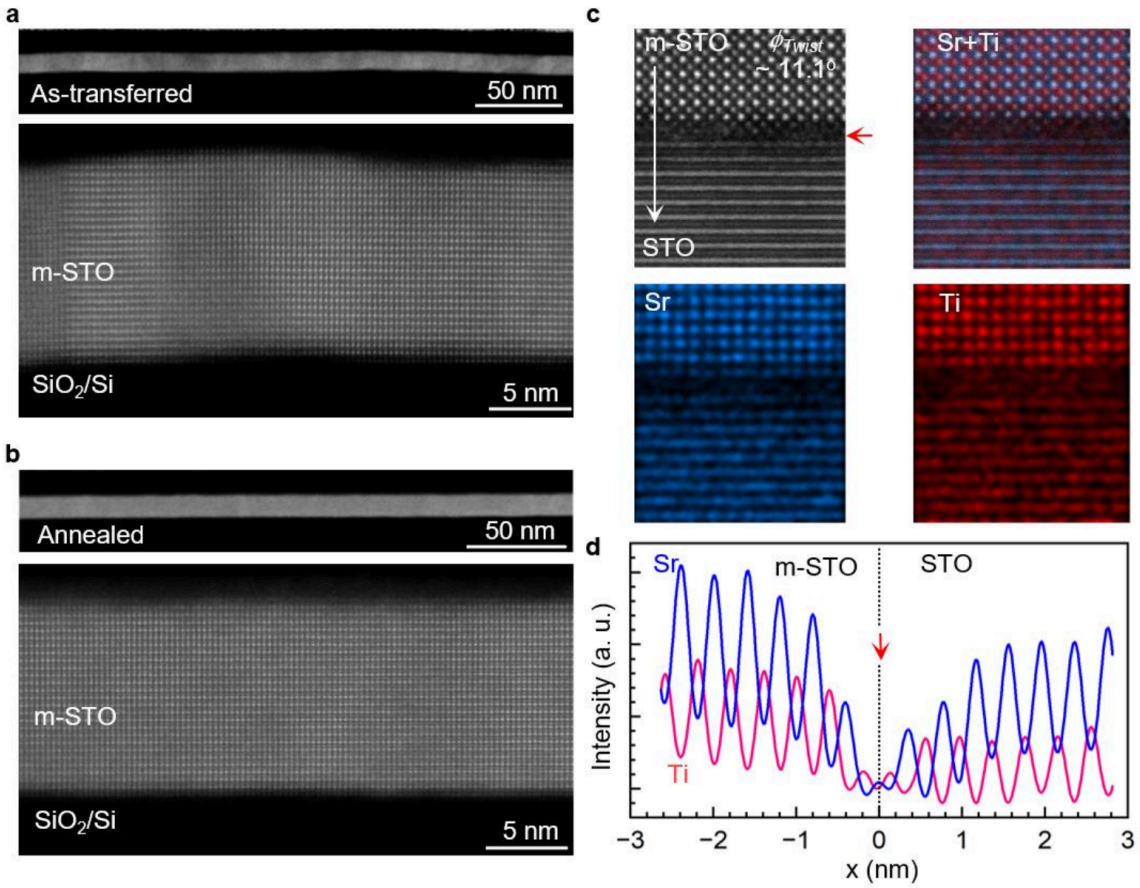
Examples

4D-STEM crystallographic phase mapping of SrHfO3



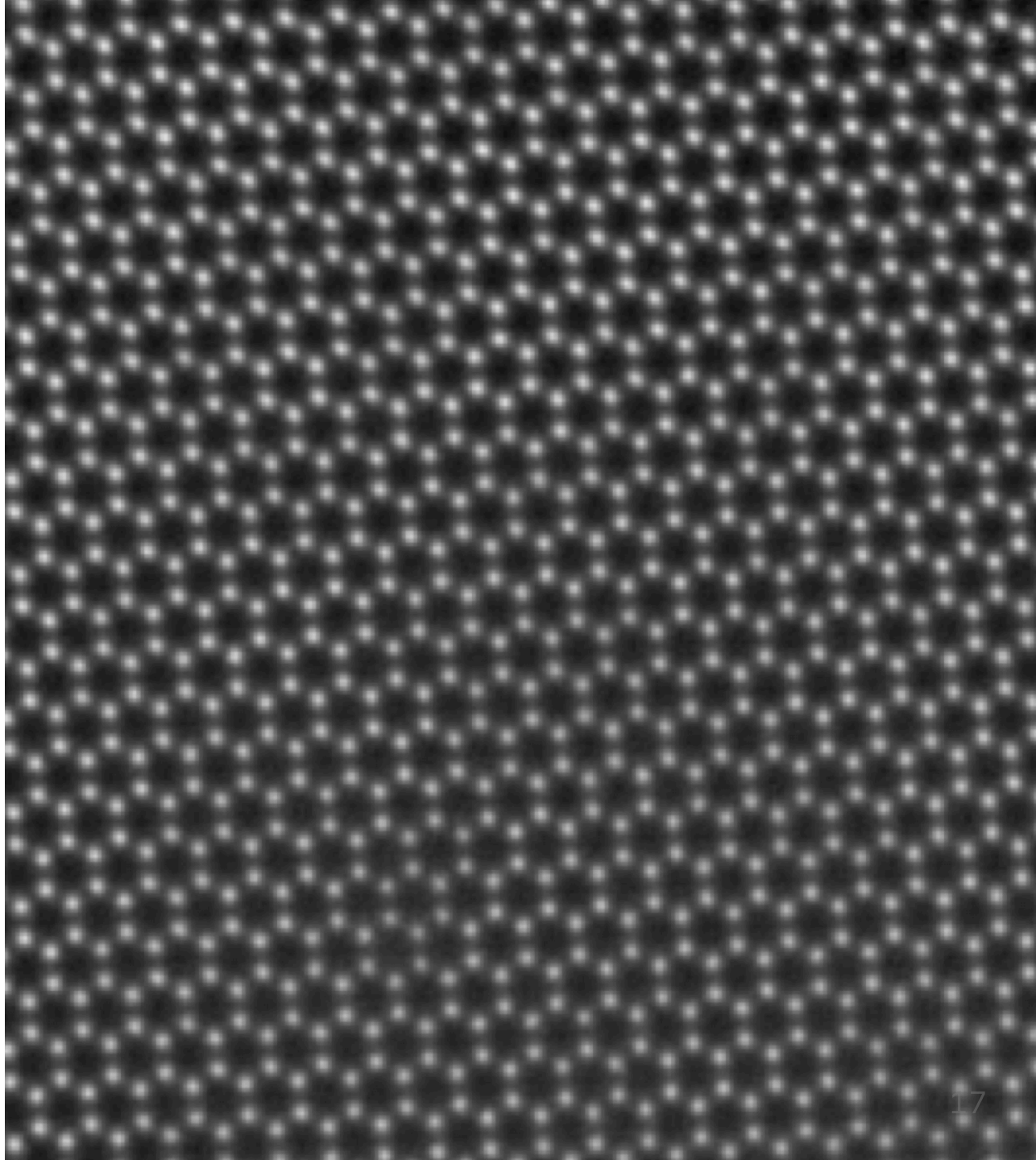
Examples

HAADF-STEM and EDS imaging of SrTiO3 interface



Examples

3D Ptychography of hBN using Themis-Z



Active topics

- Phase coexistance in high-κ materials
- Properties of high band gap oxides
- Chemical + structural analysis of oxides
- Ptychographic imaging of semiconductor heterostructure
- Ptychography algorithm improvements
- Device fabrication using perovskite oxide thin films, including FeFETs
- Strain engineering to develop novel ferroelectric perovskite oxides
- Freestanding membrane fabrication and transfer

Q & A